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Course: **Econometrics II**

1 Nonparametric estimation

Exercise 1.1. Discuss the behavior of a Nadaraya-Watson mean regression estimate when one of i.i.d. observations, say (x_1, y_1), tends to assume extreme values. Specifically, discuss

(a) the case when x_1 is very big,

(b) the case when y_1 is very big.

Solution.

(a) When x_1 is very big, we will have a big influence of this observation when estimating the regression in the right region of the support. With bounded kernels, we may find that no observations fall into the window. With unbounded kernels, the estimated regression line will go almost through (x_1, y_1).

(b) When y_1 is very big, the estimated regression line will have a hump in the region centered at x_1.

Exercise 1.2. Firms produce some product using technology $f(l,k)$. The functional form of f is unknown, although we know that it exhibits constant returns to scale. For a firm i, we observe labor l_i, capital k_i, and output y_i, and the data-generating process takes the form $y_i = f(l_i, k_i) + \varepsilon_i$, where $\mathbb{E}[\varepsilon_i] = 0$ and ε_i is independent of (l_i, k_i). Using random sample $\{y_i, k_i, l_i\}_{i=1}^n$ suggest a nonparametric estimator of $f(l,k)$ which also exhibits constant returns to scale.

Solution. The CRS technology has the property that

$$f(l,k) = kf\left(\frac{l}{k}, 1\right).$$

Hence, the data-generating process is equivalent to

$$\frac{y_i}{k_i} = f\left(\frac{l_i}{k_i}, 1\right) + \frac{\varepsilon_i}{k_i},$$

from which follows that the Nadaraya-Watson estimator can be written as

$$\hat{f}\left(\frac{l}{k}, 1\right) = \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)} \Rightarrow \hat{f}(l,k) = k \cdot \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)}.$$

Exercise 1.3. Recall the Nadaraya-Watson kernel estimator $\hat{g}(x)$ of the conditional mean $g(x) := \mathbb{E}[y|x]$ constructed for a random sample. Show that if $g(x) = c$, where c is some constant, then $\hat{g}(x)$ is unbiased.

Solution. Assume having a random sample $x_1, \ldots, x_n$, and collect it as $\mathcal{X}$. Recall that

$$\hat{g}(x) = \frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}.$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{g}(x)\right] &= \mathbb{E}\left[\mathbb{E}\left[\hat{g}(x)|\mathcal{X}\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}|\mathcal{X}\right]\right] \\
&= \mathbb{E}\left[\frac{\sum_{i=1}^n \mathbb{E}\left[y_i|\mathcal{X}\right] K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] \\
&= c\mathbb{E}\left[\frac{\sum_{i=1}^n K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] = c.
\end{aligned}$$

First equality uses the law of iterated expectations, the second uses the definition of the $\hat{g}(x)$, the third follows because kernels are functions of $\mathcal{X}$, the forth follows from the definition of the conditional mean.

Exercise 1.4. Suppose we have a random sample $\{x_i\}_{i=1}^n$ and let

$$\hat{F}(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \leq x\}$$

denote the empirical distribution function of x_i. Consider two density estimators:

- one-sided estimator:
$$\hat{f}_1(x) = \frac{\hat{F}(x+h) - \hat{F}(x)}{h},$$
- two-sided estimator:
$$\hat{f}_2(x) = \frac{\hat{F}(x+h/2) - \hat{F}(x-h/2)}{h}.$$

Show that:

(a) $\hat{F}(x)$ is an unbiased estimator of $F(x)$.

(b) The bias of $f_1(x)$ is $O(h^a)$. Find the value of a.

(c) The bias of $f_2(x)$ is $O(h^b)$. Find the value of b.

Solution. *Please, consult the note on "big-o", "small-o" notation in case derivations are unclear.*

(a) We show the unbiasedness using the fact that $\mathbb{E}\left[\mathbb{1}\{x_i \leq x\}\right] = F(x)$. Intuition is as follows: recall that $\mathbb{E}\left[\mathbb{1}\{x_i = x\}\right] = \mathbb{P}\{x_i = x\}$; intuitively, with inequality instead of equality those probabilities are summing up until the point x, which leads to equivalence with the cumulative distribution function.

Now, the unbiasedness,

$$\mathbb{E}\left[\hat{F}(x)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \leq x\}\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[\mathbb{1}\{x_i \leq x\}\right] = F(x).$$

(b) We will use the following Taylor approximation of $F(x+h)$ around x:

$$F(x+h) = F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2).$$

Substituting it, we have the bias

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_1(x) - f(x)\right] &= \frac{1}{h}\left(F(x+h) - F(x)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2) - F(x)\right) - f(x) \\
&= \frac{1}{2}hf'(x) + o(h) \Rightarrow a = 1.
\end{aligned}$$

(c) Similarly to the point (b), we expand $F\left(x+\frac{h}{2}\right)$ and $F\left(x-\frac{h}{2}\right)$ around x:

$$F\left(x+\frac{h}{2}\right) = F(x) + \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3),$$

and

$$F\left(x-\frac{h}{2}\right) = F(x) - \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3).$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_2(x)\right] - f(x) &= \frac{1}{h}\left(F\left(x+\frac{h}{2}\right) - F\left(x-\frac{h}{2}\right)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + \ldots + o(h^3) - F(x) + \ldots + o(h^3)\right) \\
&= \frac{1}{h}\left(\frac{2}{6}\left(\frac{h}{2}\right)^3 f''(x)\right) + o(h^2) \\
&= \frac{1}{24}h^2 f''(x) + o(h^2) \Rightarrow b = 2.
\end{aligned}$$

Exercise 1.5. Derive the asymptotic distribution of the Nadaraya-Watson estimator of the density of a scalar random variable x having a continuous distribution, similarly to how the asymptotic distribution of the Nadaraya-Watson estimator of the regression function is derived, under similar conditions. Give interpretation to how your expressions for asymptotic bias and asymptotic variance depend on the shape of the density.

Solution. Recall the Nadaraya-Watson density estimator, $\hat{f}(x) = \frac{1}{n}\sum_{i=1}^n K_h(x_i - x)$. During the first lecture you showed that $\mathbb{E}\left[\hat{f}(x)\right] = f(x) + o(1)$ as $h \to 0$. However, this result is not particularly revealing, since we do not see what causes the bias as h approaches (but not yet equal to) zero. To have a better picture, we work on the bias further taking a higher-order expansion of the density around x (the point we are estimating the density function at).

Now,

$$\begin{aligned}
\mathbb{E}\left[\hat{f}(x) - f(x)\right] &= \int \frac{1}{h}K\left(\frac{x_i - x}{h}\right) f(x_i)dx_i - f(x) \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u) f\left(x + hu\right) du - f(x) \\
&= \int K(u)\left(f(x) + f'(x)hu + \frac{1}{2}f''(x)h^2u^2 + o(h^2)\right) du - f(x) \\
&= f(x)\int K(u)du + f'(x)h\int uK(u)du + \frac{1}{2}f''(x)h^2\int u^2K(u)du + o(h^2) - f(x) \\
&= \frac{1}{2}f''(x)h^2\sigma_K^2 + o(h^2).
\end{aligned}$$

In particular, we use here the fact that $\int K(u)du = 1$ due to normalization of the kernel, and $\int uK(u)du = 0$ due to the kernel symmetry.

What we see here is revealing: the bias is proportional to the second derivative of the density function. The second derivative, which describes the curvature of the density, is negative ($f''(x) < 0$) around the mode of the distribution, and is positive ($f''(x) > 0$) in the tails. That means that the density estimator will be downward-biased around the mode, and upward-biased around the tails. Also note that our results do not contradict the results you have obtained during the lecture: indeed, by putting $h \to 0$, we are getting rid of the bias.

We start computing the variance with

$$\begin{aligned}
\text{var}\left[\hat{f}(x)\right] &= \text{var}\left[\frac{1}{n}\sum_{i=1}^n K_h(x_i - x)\right] = \text{var}\left[\frac{1}{n}\sum_{i=1}^n \frac{1}{h}K\left(\frac{x_i - x}{h}\right)\right] \\
&= \frac{1}{n^2h^2}\text{var}\left[\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)\right] = \frac{n}{n^2h^2}\text{var}\left[K\left(\frac{x_i - x}{h}\right)\right] = \frac{1}{nh^2}\text{var}\left[K\left(\frac{x_i - x}{h}\right)\right]
\end{aligned}$$

where the forth equality is implied by the i.i.d. assumption. Next I will normalize by nh to get

$$\begin{aligned}
nh\text{var}\left[\hat{f}(x)\right] &= \frac{1}{h}\text{var}\left[K\left(\frac{x_i-x}{h}\right)\right] \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - \frac{1}{h}\left(\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - \frac{1}{h}\left(h\cdot\frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - h\left(\mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right]\right)^2.
\end{aligned}$$

Note that we have already derived the second term (that is squared here). Specifically,

$$\mathbb{E}\left[\hat{f}(x)\right] = \mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right] = f(x) + \frac{1}{2}h^2 f''(x) + o(h^2). \tag{1}$$

Now, let's work with the first term. We have

$$\begin{aligned}
\frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] &= \frac{1}{h}\int K\left(\frac{x_i-x}{h}\right)^2 f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u)^2 f(x+hu)du \\
&= \int K(u)^2\left(f(x)+o(1)\right)du \\
&= f(x)\int K(u)^2 du + o(1) \\
&= f(x)R_K + o(1).
\end{aligned}$$

Putting everything together, we have

$$\begin{aligned}
nh\text{var}\left[\hat{f}(x)\right] &= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - h\left(\mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= f(x)R_K + o(1) - h\left(f(x)+\frac{1}{2}h^2 f''(x)+o(h^2)\right)^2 \\
&= f(x)R_K + o(1) - \left(h^2 f(x)+\frac{1}{2}h^4 f''(x)+o(h^4)\right)^2 \\
&= f(x)R_K + o(1) - \left(\mathcal{O}(h^2)\right)^2 \\
&= f(x)R_K + o(1) - \mathcal{O}(h^4) \\
&= f(x)R_K + o(1)
\end{aligned}$$

where we denote $\mathcal{O}(h^2) = h^2 f(x) + \frac{1}{2}h^4 f''(x) + o(h^4)$, because indeed by the definition of "big-o",

$$\mathcal{O}(h^2) = \frac{h^2 f(x) + \frac{1}{2}h^4 f''(x) + o(h^4)}{h^2} = f(x) + \frac{1}{2}h^2 f''(x) + o(h^2) \overset{h\to 0}{\to} f(x) < \infty.$$

Dividing $nh\text{var}\left[\hat{f}(x)\right]$ by nh, we obtain the variance of the Nadaraya-Watson density estimator,

$$\text{var}\left[\hat{f}(x)\right] = \frac{f(x)R_K}{nh} + o\left(\frac{1}{nh}\right) = \mathcal{O}\left(\frac{1}{nh}\right).$$

The CLT of the estimator is implied by the Linderberg CLT for heterogeneous random variables (for which we should verify the Linderberg condition; for simplicity we do not, but it does hold). Hence,

$$\sqrt{nh}\left(\hat{f}(x)-f(x)\right) \overset{d}{\to} \mathcal{N}\left(\frac{1}{2}\lambda f''(x), f(x)R_K\right)$$

where $\lambda := \lim_{n\to\infty} \sqrt{nh^5} < \infty \Leftrightarrow h = \mathcal{O}(n^{-1/5})$. *Notice that the convergence rate should correspond inverse-proportionally to the variance.* Remember, that $h \to 0$ in the limit so we should "delete" h from the variance to stabilize the asymptotic distribution. For example, if the convergence rate is $\sqrt{nh^2}$, we have $hf(x)R_K \to 0$ as $h \to 0$. If, on the other hand, the convergence rate is $\sqrt{nh^{-1}}$, $\frac{f(x)R_K}{h^3} \to \infty$ as $h \to 0$. The convergence rate $\sqrt{nh}$ perfectly balances those two extreme cases giving us a possibility of analyzing the non-degenerate distribution.

Exercise 1.6. Analyze carefully the asymptotic properties of the Nadaraya-Watson estimator of a regression function with pefect fit, i.e., when the variance of the error is zero.

Solution. Given the setup, we can infer that uncertainty will be coming from the distribution of x_i only, since $y_i = g(x_i)$. Let's write the Nadaraya-Watson regression estimator, plug in the data-generating process for y_i, and rewrite:

$$\hat{g}(x) = \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) y_i}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)} = \frac{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) g(x_i)}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}$$

$$\Rightarrow \hat{g}(x) - g(x) = \frac{\frac{1}{nh}\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)(g(x_i) - g(x))}{\frac{1}{nh}\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}.$$

We already know that the denominator is the Nadaraya-Watson kernel density estimator. Hence, by previous results,

$$\hat{f}(x) := \frac{1}{nh}\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right) \xrightarrow{p} f(x) \neq 0.$$

Given that the probability limit of the denominator is not zero, it is sufficient to show that the numerator is not zero either to have a tractable limiting distribution.

We now focus on the nominator, which is, however, equal to $\hat{q}_2(x)$ term that we have worked with before (during lectures). Specifically, with

$$\hat{q}_2(x) := \frac{1}{nh}\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)(g(x_i) - g(x)) \tag{2}$$

you have showed that

$$\mathbb{E}\left[\hat{q}_2(x)\right] = h^2 B(x) f(x) \sigma_K^2 + o(h^2), \quad \text{var}\ \left[\hat{q}_2(x)\right] = o\left(\frac{1}{nh}\right),$$

with $B(x) := \frac{g''(x)}{2} + \frac{g'(x)f'(x)}{f(x)}$. Equation (2) is the average of random variables

$$\zeta_i(x) := \frac{1}{h}K\left(\frac{x_i - x}{h}\right)(g(x_i) - g(x)).$$

We now focus on the expectation and the variance of $\zeta_i(x)$ to derive its distribution, and once we have it, we could derive the distribution of $\hat{g}(x)$ as well.

Now,

$$\begin{aligned}
\mathbb{E}\left[\zeta_i(x)^2\right] &= \frac{1}{h^2}\int K\left(\frac{x_i-x}{h}\right)^2 (g(x_i)-g(x))^2 f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \frac{1}{h}\int K(u)^2 \left(g(x+hu)-g(x)\right)^2 f(x+hu)du \\
&= \frac{1}{h}\int K(u)^2 \left(g'(x)hu+o(h)\right)^2 \left(f(x)+o(1)\right)du \\
&= \frac{1}{h}\int \left(g'(x)hu+o(h)\right)^2 K(u)^2 f(x)du + o(h) \\
&= hg'(x)^2 f(x)\int u^2 K(u)^2 du + o(h) \\
&= hg'(x)^2 f(x)\Psi_K^2 + o(h)
\end{aligned}$$

with $\Psi_K^2 := \int u^2 K(u)^2 du$. Similarly,

$$\begin{aligned}
\mathbb{E}\left[\zeta_i(x)\right] &= \int \frac{1}{h}K\left(\frac{x_i-x}{h}\right)\left(g(x_i)-g(x)\right)f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u)\left(g(x+hu)-g(x)\right)f(x+hu)du \\
&= \int K(u)\cdot o(1)(f(x)+f'(x)hu)du = o(h).
\end{aligned}$$

Together, the variance is $\operatorname{var}\left[\zeta_i(x)\right] = \mathbb{E}\left[\zeta_i(x)^2\right] - \mathbb{E}[\zeta_i(x)]^2$ or

$$hg'(x)^2 f(x)\Psi_K^2 + o(h) + o(h^2) = hg'(x)^2 f(x)\Psi_K^2 + o(h).$$

By some form of the CLT (notice the change of the convergence rate), we obtain the limiting distribution of ζ_i,

$$\sqrt{nh^{-1}}\left(\frac{1}{nh}\sum_{i=1}^{n}K\left(\frac{x_i-x}{h}\right)\left(g(x_i)-g(x)\right) - h^2 B(x)\sigma_K^2 f(x) + o(h^2)\right) \xrightarrow{d} \mathcal{N}\left(0, g'(x)^2 f(x)\Psi_K^2\right).$$

We are finally ready to derive the limiting distribution of $\hat{g}(x)$. So by the CLT and Slustky's theorem,

$$\sqrt{nh^{-1}}(\hat{g}(x)-g(x)) \xrightarrow{d} \mathcal{N}\left(\phi\sigma_K^2 B(x), \frac{g'(x)^2}{f(x)}\Psi_K^2\right)$$

with $\phi := \lim_{n\to\infty}\sqrt{nh^3}$.

Exercise 1.7. This problem illustrates the use of a difference operator in nonparametric estimation with i.i.d. data. Suppose there is a scalar variable z that takes values on a bounded support. For simplicity, let z be deterministic and compose a uniform grid on the unit interval $[0,1]$. The other variables are i.i.d. Assume that for the function $g(\cdot)$ below the following Lipschitz condition is satisfied:

$$|g(u)-g(v)| \leq G\,|u-v|$$

for some constant G.

1. Consider a nonparametric regression of y on z:

   $$y_i = g(z_i) + e_i, \quad i = 1,\ldots,n,$$

   where $\mathbb{E}[e_i|z_i] = 0$. Let the data $\{(z_i, y_i)\}_{i=1}^n$ be ordered so that the zs are in increasing order. A first difference transformation results in the following set of equations:

   $$y_i - y_{i-1} = g(z_i) - g(z_{i-1}) + e_i - e_{i-1}, \quad i = 2,\ldots,n. \tag{3}$$

   The target is to estimate $\sigma^2 := \mathbb{E}\left[e_i^2\right]$. Propose its consistent estimator based on the first-difference transformed regression (8). Prove consistency of your estimator.

2. Consider the following partially linear regression of y on x and z:

   $$y_i = x_i'\beta + g(z_i) + e_i, \quad i = 1,\ldots,n, \tag{4}$$

   where $\mathbb{E}\left[e_i|x_i, z_i\right] = 0$. Let the data $\{x_i, z_i, y_i\}_{i=1}^n$ be ordered so that zs are in increasing order. The target is to nonparametrically estimate $g(z)$. Propose its consistent estimator using the first-difference transformation of (4).

Solution.

1. Start with the squared average,

$$\begin{aligned}\frac{1}{n-1}\sum_{i=2}^{n}(y_i-y_{i-1})^2 = &\frac{1}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))^2+\frac{1}{n-1}\sum_{i=2}^{n}(e_i-e_{i-1})^2\\ &-\frac{2}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))(e_i-e_{i-1}).\end{aligned} \tag{5}$$

The logic behind the estimator (5) is that we hope terms with $g(z_i)-g(z_{i-1})$ will disappear due to the smoothness of the function (that asymptotically the difference will converge to zero).

Since z_i compose a uniform grid and are in increasing order, we know that $z_i - z_{i-1} = \frac{1}{n-1}$, and we can find the limit of the first term using the Lipschitz condition as

$$\begin{aligned}\left|\frac{1}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))^2\right| &\le \frac{G^2}{n-1}\sum_{i=2}^{n}(z_i-z_{i-1})^2\\ &=\frac{G^2}{n-1}\frac{n-1}{(n-1)^2}=\frac{G^2}{(n-1)^2}\to 0\end{aligned}$$

as $n\to\infty$. We can similarly use the Lipschitz condition for the third term,

$$\begin{aligned}\left|\frac{2}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))(e_i-e_{i-1})\right| &\le \frac{2G}{n-1}\sum_{i=2}^{n}|z_i-z_{i-1}|\,|e_i-e_{i-1}|\\ &=\frac{2G}{(n-1)^2}\sum_{i=2}^{n}|e_i-e_{i-1}|\\ &\le\frac{2G}{(n-1)^2}\sum_{i=2}^{n}(|e_i|+|e_{i+1}|)\\ &=\frac{2G}{n-1}\frac{1}{n-1}\sum_{i=2}^{n}(|e_i|+|e_{i-1}|)\xrightarrow{p}\frac{2G}{n-1}\cdot 2\mathbb{E}\left[|e_i|\right]\to 0\end{aligned}$$

as $n\to\infty$. The first inequality uses the Lipschitz condition of the function, the second inequality uses the triangular inequality, and the probability limit is followed by the law of large numbers and i.i.d. data. *Note that* $\mathbb{E}[|e_i|]\neq 0$*!* We assume that $\mathbb{E}[e_i|z_i]=0$, but the absolute value takes into account only positive values of the random variable.

The second term results in the variance we are after:

$$\frac{1}{n-1}\sum_{i=2}^{n}(e_i-e_{i-1})^2=\frac{1}{n-1}\sum_{i=2}^{n}e_i^2-2e_ie_{i-1}+e_{i-1}^2\xrightarrow{p}2\mathbb{E}[e_i^2]=2\sigma.$$

Hence, the consistent estimator of the variance is

$$\hat\sigma^2=\frac{1}{2}\frac{1}{n-1}\sum_{i=2}^{n}(y_i-y_{i-1})^2.$$

2. From the first-difference regression we have

$$\Delta y_i=\Delta x_i'\beta+\Delta g(z_i)+\Delta e_i$$

where Δ denotes the difference parameter. Consider an estimator of β,

$$\hat\beta=\left(\sum_{i=2}^{n}\Delta x_i\Delta x_i'\right)^{-1}\sum_{i=2}^{n}\Delta x_i\Delta y_i. \tag{6}$$

Logic behind (6) is very similar to the one in the previous point: we take the difference and hope that the difference between $g(\cdot)$ function at neighboring points will converge to zero.

To prove consistency formally, we have

$$\hat{\beta} - \beta = \left(\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta x_i' \right)^{-1} \left(\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \left(\Delta g(z_i) + \Delta e_i \right) \right).$$

The denominator is equal in the limit to some non-zero constant C by the law of large numbers,

$$\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta x_i' \xrightarrow{p} C.$$

The product

$$\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta e_i \xrightarrow{p} 0$$

due to the law of iterated expectations and $\mathbb{E}[e_i | x_i, z_i] = 0$. The remaining term,

$$\begin{aligned} \left| \frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta g(z_i) \right| &= \left| \frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \left(g(z_i) - g(z_{i-1}) \right) \right| \\ &\leq \frac{G}{n-1} \sum_{i=2}^{n} |\Delta x_i| |z_i - z_{i-1}| \\ &= \frac{G}{n-1} \frac{1}{n-1} \sum_{i=2}^{n} |\Delta x_i| \xrightarrow{p} 0 \end{aligned}$$

as $n \to 0$, where we again use the Lipschitz condition. As a result, $\hat{\beta} - \beta \xrightarrow{p} 0$.

Having found a consistent estimator of β, we can apply standard nonparametrics to the following regression:

$$y_i - x_i' \hat{\beta} = g(z_i) + e_i^\star, \quad e_i^\star := e_i + x_i'(\beta - \hat{\beta}).$$

Note that if $\hat{\beta}$ was not consistent, we would be working with a misspecified model since $\mathbb{E}[e_i^\star | x_i, z_i] \neq 0$.

Consider the following estimator (we select the rectangular kernel for simplicity),

$$\begin{aligned} \hat{g}(z) &= \frac{\sum_{i=1}^{n} \left(y_i - x_i' \hat{\beta} \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} \\ &= \frac{\sum_{i=1}^{n} \left(g(z_i) + e_i + x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\} \right)}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} \\ &= \frac{\sum_{i=1}^{n} g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} + \frac{\sum_{i=1}^{n} e_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} + \frac{\sum_{i=1}^{n} x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} \end{aligned}$$

We work with the first term now. Notice that

$$\left| \frac{\sum_{i=1}^{n} \left(g(z_i) - g(z) \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} \right| \leq G \left| \frac{\sum_{i=1}^{n} |z_i - z| \, \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} \right| \leq Gh \tag{7}$$

which holds because all differences $|z_i - z|$ are less than h due to the indicator function, so we bound the whole expression by h. From it follows that

$$\frac{\sum_{i=1}^{n} g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} = \frac{\sum_{i=1}^{n} \left(g(z) + g(z_i) - g(z) \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{ |z_i - z| \leq h \right\}} \leq g(z) + Gh \to g(z)$$

as $h \to 0$ where we use (7).

For the third term, it holds by the law of iterated expectations and of large numbers that

$$\frac{\sum_{i=1}^{n} x_i' \mathbb{1}\{|z_i - z| \le h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \le h\}} \xrightarrow{p} \frac{\mathbb{E}\left[x_i' \mathbb{1}\{|z_i - z| \le h\}\right]}{\mathbb{E}\left[\mathbb{1}\{|z_i - z| \le h\}\right]} = \mathbb{E}[x_i'],$$

and by $\hat{\beta} - \beta \xrightarrow{p} 0$, we have that

$$\frac{\sum_{i=1}^{n} x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \le h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \le h\}} \xrightarrow{p} 0.$$

Finally, for the third term, we have by the law of iterated expectations

$$\frac{\sum_{i=1}^{n} e_i \mathbb{1}\{|z_i - z| \le h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \le h\}} \xrightarrow{p} 0.$$

Hence, $\hat{g}(z) \xrightarrow{p} g(z)$.

Exercise 1.8. Let $(x_i, y_i)_{i=1}^n$ be a random sample from the population of (x, y), where x has a discrete distribution with the support $a_{(1)}, \ldots, a_{(k)}$, where $a_{(1)} < \ldots < a_{(k)}$. Having written the conditional expectation $\mathbb{E}[y|x = a_{(j)}]$ in the form that allows to apply the analogy principle, propose an analog estimator $\hat{g}_j$ of $g_j = \mathbb{E}[y|x = a_{(j)}]$ and derive its asymptotic distribution.

Solution. Fix $a_{(j)}$ for $j = 1, \ldots, k$. Then

$$g_j := g_i(a_{(j)}) = \mathbb{E}[y_i | x_i = a_{(j)}] = \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]},$$

which holds because

$$\begin{aligned}
\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\} | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\
&\quad + \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\} | x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\
&= \mathbb{P}\{x_i = a_{(j)}\},
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\} | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\
&\quad + \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\} | x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\
&= \mathbb{E}[y_i | x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\}.
\end{aligned}$$

Hence, the estimator is

$$\hat{g}_i(a_{(j)}) = \frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \xrightarrow{p} \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]} := g_i(a_{(j)}).$$

To derive the asymptotic distribution, normalize

$$\begin{aligned}
\sqrt{n}(\hat{g}_j - g_j) &= \sqrt{n} \left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} - g_j \right) \\
&= \sqrt{n} \left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\} - g_j \sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \right) \\
&= \sqrt{n} \left(\frac{\sum_{i=1}^n (y_i - \mathbb{E}[y_i | x_i = a_{(j)}]) \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \right).
\end{aligned}$$

So by the CLT, the numerator is distributed as

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])\mathbb{1}\{x_i = a_{(j)}\} \xrightarrow{d} \mathcal{N}(0, \omega),$$

where ω is the asymptotic variance given by

$$\begin{aligned}\omega &:= \text{var}\left[(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])\mathbb{1}\{x_i = a_{(j)}\}\right] \\ &= \mathbb{E}[(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])^2|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\ &= \text{var}\left[y_i|x_i = a_{(j)}\right] \cdot \mathbb{P}\{x_i = a_{(j)}\}.\end{aligned}$$

By the Slutsky's theorem and CMT, we have that

$$\sqrt{n}(\hat{g}_j - g_j) \xrightarrow{d} \mathcal{N}\left(0, \frac{\text{var}\left[y_i|x_i = a_{(j)}\right]}{\mathbb{P}\{x_i = a_{(j)}\}}\right).$$

2 Discrete choice models

Exercise 2.1. Let X be distibuted as $\Lambda(x) := \dfrac{e^x}{1+e^x}$. Verify that

a) $\dfrac{\partial \Lambda(x)}{\partial x} = \Lambda(x)(1-\Lambda(x))$,

b) $\dfrac{\partial \log \Lambda(x)}{\partial x} = 1 - \Lambda(x)$,

c) $-\dfrac{\partial^2 \log \Lambda(x)}{\partial x^2} = \Lambda(x)(1-\Lambda(x))$.

Solution.

a) Using the usual rule from calculus,

$$\frac{\partial \Lambda(x)}{\partial x} = \frac{e^x(1+e^x) - e^x e^x}{(1+e^x)^2} = \frac{e^x}{1+e^x} \cdot \frac{1}{1+e^x} = \Lambda(x)(1-\Lambda(x)).$$

b) First, $\log \Lambda(x) = \log\left(\frac{e^x}{1+e^x}\right) = x - \log(1+e^x)$, then

$$\frac{\partial \log \Lambda(x)}{\partial x} = 1 - \frac{e^x}{1+e^x} = 1 - \Lambda(x).$$

c) The result immediately follows from a) and b).

Lemma 2.2. *For some constant a, we have*

$$\int_{-\infty}^{\infty} a e^{-\varepsilon} \exp\left(-a e^{-\varepsilon}\right) d\varepsilon = 1.$$

Proof. Substitute $u = ae^{-\varepsilon}$. Then $\varepsilon = -\log u + \log a$, and $d\varepsilon = -(1/u)du$. We change the bounds of the integral as $\varepsilon \to -\infty \Rightarrow u \to \infty$, and $\varepsilon \to \infty \Rightarrow u \to 0$. Hence

$$-\int_{\infty}^{0} u \exp(-u) \cdot \frac{1}{u} du = -\int_{\infty}^{0} \exp(-u) du = \int_{0}^{\infty} \exp(-u) du = 1.$$

This completes the proof. □

Exercise 2.3. Derive the logit from the Gumbel distribution of ε_0 and ε_1.

Solution. We start with

$$\begin{aligned}
\mathbb{P}\{y=1\} &= \mathbb{P}\{\varepsilon_0 < \varepsilon_1 + V_1 - V_0\} \\
&= F_{\varepsilon_0}\left(\varepsilon_1 + V_1 - V_0\right) \\
&= \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\varepsilon_1+V_1-V_0} f(\varepsilon_0,\varepsilon_1)d\varepsilon_0\right)d\varepsilon_1 \\
&= \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\varepsilon_1+V_1-V_0} f(\varepsilon_0)f(\varepsilon_1)d\varepsilon_0\right)d\varepsilon_1
\end{aligned}$$

due to independence of ε_0 and ε_1, so

$$\mathbb{P}\{y=1\} = \int_{-\infty}^{\infty} f(\varepsilon_1)\left(\int_{-\infty}^{\varepsilon_1+V_1-V_0} f(\varepsilon_0)d\varepsilon_0\right)d\varepsilon_1. \tag{8}$$

By our assumption,

$$f(\varepsilon_i) = e^{-\varepsilon_i}\exp\left(-e^{-\varepsilon_i}\right), \quad i=1,2.$$

Plugging it in (8), we have

$$\begin{aligned}
\mathbb{P}\{y=1\} &= \int_{-\infty}^{\infty} f(\varepsilon_1)\left(\int_{-\infty}^{\varepsilon_1+V_1-V_0} e^{-\varepsilon_0}\exp\left(-e^{-\varepsilon_0}\right)d\varepsilon_0\right)d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f(\varepsilon_1)\left[\exp\left(-e^{-\varepsilon_0}\right)\right]_{-\infty}^{\varepsilon_1+V_1-V_0} d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} f(\varepsilon_1)\exp\left(-e^{-(\varepsilon_1+V_1-V_0)}\right)d\varepsilon_1.
\end{aligned}$$

Now, plug in the density for $f(\varepsilon_1)$ to obtain

$$\begin{aligned}
\mathbb{P}\{y=1\} &= \int_{-\infty}^{\infty} e^{-\varepsilon_1}\exp\left(-e^{-\varepsilon_1}\right)\exp\left(-e^{-(\varepsilon_1+V_1-V_0)}\right)d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}\exp\left(\left(-e^{-\varepsilon_1} - e^{-(\varepsilon_1+V_1-V_0)}\right)\right)d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}\exp\left(\left(-e^{-\varepsilon_1} - e^{-\varepsilon_1}e^{-(V_1-V_0)}\right)\right)d\varepsilon_1 \\
&= \int_{-\infty}^{\infty} e^{-\varepsilon_1}\exp\left(-e^{-\varepsilon_1}\left(1+e^{-(V_1-V_0)}\right)\right)d\varepsilon_1.
\end{aligned}$$

This integral is equal to

$$\mathbb{P}\{y=1\} = \frac{1}{1+e^{-(V_1-V_0)}} = \frac{e^{V_1}}{e^{V_0}+e^{V_1}} = \frac{e^{V_1-V_0}}{1+e^{V_1-V_0}} = \Lambda\left(V_1-V_0\right)$$

due to the lemma above with $a := 1+e^{-(V_1-V_0)}$.

Exercise 2.4. Consider a latent variable modeled by $y_i^\star = x_i'\beta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$. Suppose we observe only $y_i = 1$ if $y_i^\star < U_i$, and $y_i = 0$ if $y_i \geq U_i$, where the upper limit U_i is a known constant for each i and may differ over i.

a) Find $\mathbb{P}\{y_i = 1|x_i\}$.

b) Provide details on how to estimate β with maximum likelihood.

Solution.

a) Similarly as in the lecture,

$$\begin{aligned}
\mathbb{P}\{y_i=1\} &= \mathbb{P}\{y_i^\star < U_i\} \\
&= \mathbb{P}\{x_i'\beta + \varepsilon_i < U_i\} \\
&= \Phi\left(U_i - x_i'\beta\right).
\end{aligned}$$

b) Conditional likelihood is $\mathcal{L} = p^y(1-p)^{1-y}$ with $p = \Phi\left(U_i - x_i'\beta\right)$. Then,

$$\log \mathcal{L} = \sum_{i=1}^{n} y_i \log \Phi\left(U_i - x_i'\beta\right) + (1-y_i)\log\left(1 - \Phi\left(U_i - x_i'\beta\right)\right).$$

Taking the first-order condition w.r.t. β,

$$\begin{aligned}
\frac{\partial \log \mathcal{L}}{\partial \beta} &= \sum_{i=1}^{n} -\frac{y_i}{\Phi\left(U_i - x_i'\beta\right)}\phi\left(U_i - x_i'\beta\right)x_i' + \frac{1-y_i}{1-\Phi\left(U_i - x_i'\beta\right)}\phi\left(U_i - x_i'\beta\right)x_i' \\
&= \sum_{i=1}^{n} \frac{\Phi\left(U_i - x_i'\beta\right) - y_i}{\Phi\left(U_i - x_i'\beta\right)\left(1 - \Phi\left(U_i - x_i'\beta\right)\right)}\phi\left(U_i - x_i'\beta\right)x_i' = 0.
\end{aligned}$$

Solving for β gives us the ML estimator of β.

Exercise 2.5. In 2022, CERGE-EI conducted a small research aimed at studying a relationship between drinking alcohol and the pass rate of the Econometrics II course. Table 1 presents the number of students in each category. Dummy variable d_i takes a value of 1 if a student i has been drinking one drink per day during the final week, 0 otherwise.

Table 1: Data on 100 students

$d_i \setminus y_i$	fail	pass
no beer	24	28
beer	32	16

To answer this research question, CERGE-EI committee decided to specify the model as

$$y_i^\star = \alpha + \beta d_i + \varepsilon_i, \quad i = 1, \ldots, 100,$$

and they observe only

$$y_i = \begin{cases} 1 & \text{if } y_i^\star > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Obtain the maximum-likelihood estimators of α and β specifying the model as logit. Should you drink during the final week?

Solution. First, we write the log-likelihood function as

$$\begin{aligned}
\log \mathcal{L}(\alpha, \beta) = &\sum_{0,0} \log \Pr\{y_i = 0, d_i = 0\} + \sum_{0,1} \log \Pr\{y_i = 0, d_i = 1\} \\
&+ \sum_{1,0} \log \Pr\{y_i = 1, d_i = 0\} + \sum_{1,1} \log \Pr\{y_i = 1, d_i = 1\}
\end{aligned}$$

where $\sum_{i,j}$ is a summation of i that have $y_i = i$ and $d_i = j$. We calculate probabilities using the model assumptions:

- $\Pr\{y_i = 0, d_i = 0\} = \Pr\{\alpha + \varepsilon \leq 0\} = F(-\alpha) = 1 - F(\alpha)$,
- $\Pr\{y_i = 0, d_i = 1\} = \Pr\{\alpha + \beta + \varepsilon \leq 0\} = F(-\alpha - \beta) = 1 - F(\alpha + \beta)$,
- $\Pr\{y_i = 1, d_i = 0\} = \Pr\{\alpha + \varepsilon > 0\} = 1 - F(-\alpha) = F(\alpha)$,
- $\Pr\{y_i = 1, d_i = 1\} = \Pr\{\alpha + \beta + \varepsilon > 0\} = 1 - F(-\alpha - \beta) = F(\alpha + \beta)$.

Now we simplify the log-likelihood as follows:

$$\log \mathcal{L}(\alpha, \beta) = 24 \log(1 - F(\alpha)) + 32 \log(1 - F(\alpha + \beta)) + 28 \log F(\alpha) + 16 \log F(\alpha + \beta).$$

Taking the first-order condition w.r.t. α,

$$\frac{\partial \log \mathcal{L}(\alpha, \beta)}{\partial \alpha} = -\frac{24}{1 - F(\alpha)}F'(\alpha) - \frac{32}{1 - F(\alpha + \beta)}F'(\alpha + \beta) + \frac{28}{F(\alpha)}F'(\alpha) + \frac{16}{F(\alpha + \beta)}F'(\alpha + \beta) = 0,$$

and w.r.t. β,

$$\frac{\partial \log \mathcal{L}(\alpha, \beta)}{\partial \beta} = -\frac{32}{1 - F(\alpha + \beta)} F'(\alpha + \beta) + \frac{16}{F(\alpha + \beta)} F'(\alpha + \beta) = 0.$$

Substituting the second equation into the first one, the first one reduces to

$$\frac{24}{1 - F(\alpha)} F'(\alpha) = \frac{28}{F(\alpha)} F'(\alpha),$$

from which follows that $F(\alpha) = 28/52$.

Analogously, from the second equation,

$$\frac{32}{1 - F(\alpha + \beta)} F'(\alpha + \beta) = \frac{16}{F(\alpha + \beta)} F'(\alpha + \beta),$$

from which follows that $F(\alpha + \beta) = 16/48$.

Now, substitute for $F(x) = \dfrac{e^x}{1 + e^x}$ to have

$$F(\alpha) = \frac{e^{\hat{\alpha}}}{1 + e^{\hat{\alpha}}} = \frac{28}{52}.$$

By simple calculations, $\hat{\alpha} = 0.156$. Similarly,

$$F(\alpha + \beta) = \frac{e^{\hat{\alpha} + \hat{\beta}}}{1 + e^{\hat{\alpha} + \hat{\beta}}} = \frac{16}{48}.$$

By simple calculations, $\hat{\beta} = -0.850$. There is a clear negative effect of drinking on the pass rate.

Exercise 2.6. For the conditional logit,

$$p_{ij} = \frac{e^{x'_{ij}\beta}}{\sum\limits_{\ell} e^{x'_{i\ell}\beta}},$$

show that

a) $\dfrac{\partial p_{ij}}{\partial \beta} = p_{ij}(x_{ij} - \bar{x}_i)$, where $\bar{x}_i := \sum\limits_{l=1}^{m} p_{i\ell} x_{i\ell}$, and

b) $\dfrac{\partial p_{ij}}{\partial x_{ik}} = p_{ij}(\delta_{ijk} - p_{ik})\beta$, where $\delta_{ijk} := \mathbb{1}\{j = k\}$.

Solution.

a) Differentiating by parts,

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \beta} &= \frac{e^{x'_{ij}\beta} \cdot x_{ij} \cdot \sum_{\ell} e^{x'_{i\ell}\beta} - e^{x'_{ij}\beta} \cdot \sum_{\ell} e^{x'_{i\ell}\beta} \cdot x_{i\ell}}{\left(\sum_{\ell} e^{x'_{i\ell}\beta}\right)^2} \\
&= \frac{e^{x'_{ij}\beta}}{\sum_{\ell} e^{x'_{i\ell}\beta}} x_{ij} - \frac{e^{x'_{ij}\beta}}{\left(\sum_{\ell} e^{x'_{i\ell}\beta}\right)^2} \sum_{\ell} e^{x'_{i\ell}\beta} \cdot x_{i\ell} \\
&= p_{ij} x_{ij} - p_{ij} \sum_{\ell} p_{i\ell} x_{i\ell} = p_{ij}(x_{ij} - \bar{x}_i).
\end{aligned}$$

b) For the marginal effect, we differentiate between $j = k$ and $j \neq k$. For the first case,

$$\frac{\partial p_{ij}}{\partial x_{ij}} = \frac{e^{x'_{ij}\beta} \cdot \beta}{\sum_{\ell} e^{x_{i\ell}\beta}} - \frac{e^{x'_{ij}\beta}}{\left(\sum_{\ell} e^{x'_{i\ell}\beta}\right)^2} e^{x'_{ij}\beta} \cdot \beta = p_{ij}(1 - p_{ij})\beta,$$

and for the second case,

$$\frac{\partial p_{ij}}{\partial x_{ik}} = -\frac{e^{x'_{ij}\beta}}{\left(\sum_\ell e^{x'_{i\ell}\beta}\right)^2} \cdot e^{x'_{ik}\beta} \cdot \beta = -p_{ij}p_{ik}\beta,$$

so altogether,

$$\frac{\partial p_{ij}}{\partial x_{ij}} = p_{ij}(\delta_{ijk} - p_{ik})\beta.$$

Exercise 2.7. You have 20 observations that are drawn from a censored normal distribution such that 6 are equal to zero. The applicable model is

$$y_i^\star = \mu + \varepsilon_i, \quad y_i = \begin{cases} y_i^\star & \text{if } \mu + \varepsilon_i > 0 \\ 0 & \text{otherwise} \end{cases}, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

a) Discuss the OLS estimator $\hat{\mu}$ of μ. Do you expect this estimator to over- or underestimate μ?

b) If we consider only the nonzero observations, then the truncated regression model applies. Discuss the OLS estimator $\tilde{\mu}$ of μ in this case. Do you expect this estimator to over- or underestimate μ? Is $\hat{\mu} > \tilde{\mu}$?

c) Consider the tobit model that applies to the full data set. Formulate the log-likelihood. Reformulate the log-likelihood in terms of $\theta := 1/\sigma$ and $\gamma := \mu/\sigma$. Derive the necessary conditions for maximizing the log-likelihood with respect to θ and γ.

d) Repeat c) in the context of the truncated regression model.

Solution.

a) The OLS estimator in this case is just a sample mean. Because all negative values have been transformed into zeroes, the estimator $\hat{\mu}$ would overestimate the true mean.

b) Likewise, for the truncated mean $\tilde{\mu}$, whereas a complete sample might include some negative values, the observed one does not. Once again, this will serve to inflate the estimator of the mean. Because the censored mean includes zeros, we have $\tilde{\mu} > \hat{\mu}$ (see the following Figure 1 where censored (green) and truncated (blue) means are computed for 1000 iterations with $\mu = 2$).

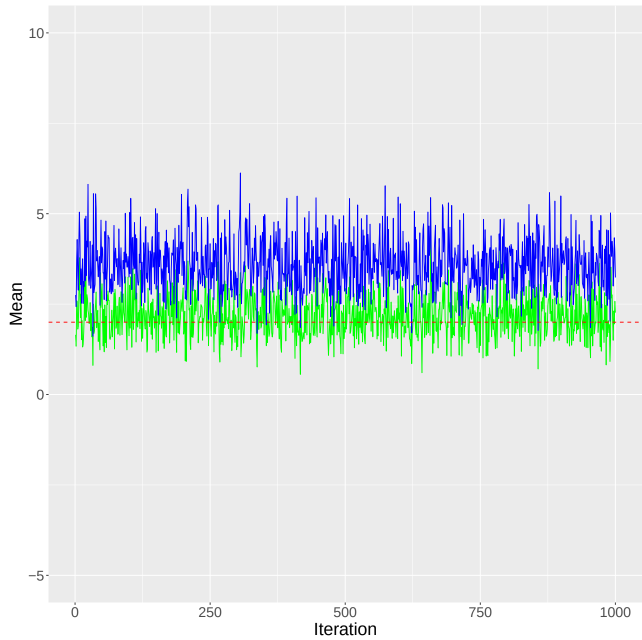

Figure 1: Censored and truncated mean

c) Denote the number of observations which are non-zero as n_1, and the number of zero (censored) observations as n_0. Then, $n = n_0 + n_1$. The log-likelihood for the tobit model is then

$$\log \mathcal{L}(y_i; \mu, \sigma^2) = -\frac{n_1}{2}\left(\log 2\pi + \log \sigma^2\right) - \frac{1}{2\sigma^2}\sum_{n_1}(y_i - \mu)^2 + \sum_{n_0}\log\left(1 - \Phi\left(\frac{\mu}{\sigma}\right)\right),$$

and with $\theta := 1/\sigma$ and $\gamma := \mu/\sigma$,

$$\log \mathcal{L}(y_i; \theta, \gamma) = -\frac{n_1}{2}\left(\log 2\pi + \log \frac{1}{\theta^2}\right) - \frac{1}{2}\sum_{n_1}(\theta y_i - \gamma)^2 + \sum_{n_0}\log\left(1 - \Phi\left(\gamma\right)\right).$$

The first-order conditions are

$$\begin{aligned}\frac{\partial \mathcal{L}(y_i; \theta, \gamma)}{\partial \gamma} &= \sum_{n_1}(\theta y_i - \gamma) - \sum_{n_0}\frac{\phi(\gamma)}{1 - \Phi(\gamma)} \\ &= \sum_{n_1}\theta y_i - n_1\gamma - n_0\frac{\phi(\gamma)}{1 - \Phi(\gamma)} = 0,\end{aligned}$$

and

$$\frac{\partial \mathcal{L}(y_i; \theta, \gamma)}{\partial \theta} = \frac{n_1}{\theta} - \sum_{n_1}(\theta y_i - \gamma)y_i = 0.$$

d) In the truncated case, the log-likelihood for the tobit model is

$$\log \mathcal{L}(y_i; \mu, \sigma^2) = -\frac{n_1}{2}\left(\log 2\pi + \log \sigma^2\right) - \frac{1}{2\sigma^2}\sum_{n_1}(y_i - \mu)^2 - \sum_{n_1}\log\Phi\left(\frac{\mu}{\sigma}\right),$$

or transformed,

$$\log \mathcal{L}(y_i; \theta, \gamma) = -\frac{n_1}{2}\left(\log 2\pi + \log \frac{1}{\theta^2}\right) - \frac{1}{2}\sum_{n_1}(\theta y_i - \gamma)^2 - \sum_{n_1}\log\Phi\left(\gamma\right).$$

Taking the first-order conditions,

$$\frac{\partial \log \mathcal{L}(y_i; \theta, \gamma)}{\partial \gamma} = \sum_{n_1}(\theta y_i - \gamma) - n_1\frac{\phi(\gamma)}{\Phi(\gamma)} = 0,$$

and

$$\frac{\partial \log \mathcal{L}(y_i; \theta, \gamma)}{\partial \theta} = \frac{n}{\theta} - \sum_{n_1}(\theta y_i - \gamma)y_i = 0.$$

Exercise 2.8. We have data for $n = 753$ married women aged 30-60 in the US in 1975. 428 women in the sample work, and 325 do not work.

a) For the sample of working women, the following equation was estimated in *Stata* by the OLS:

$$\log(\text{wage})_i = \beta_0 + \beta_1\text{age}_i + \beta_2\text{educ}_i + \beta_3\text{exper}_i + \beta_4\text{expersq}_i + u_i$$

with the following results.

Is the OLS estimator biased?

b) Suppose that you have the following variables that can be used as exclusion restrictions for the Heckman selection correction:

- `kids05`: the number of kids between 0 and 5,
- `kids618`: the number of kids between 6 and 18,
- `nwifeinc`: household income excluding wife's income.

Describe how would you implement it.

c) The Heckman selection correction was implemented using `kids05`, `kids618`, and `nwifeinc` as exclusion restrictions with the following results:

Comment on the sign of the parameter for the inverse Mills ratio (`imr`). Is there evidence for the selection bias?

Linear regression						
				Number of obs	=	428
				F(4, 423)	=	20.42
				Prob > F	=	0.0000
				R-squared	=	0.1568
				Root MSE	=	.6672

lwage	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
age	.0002836	.0054379	0.05	0.958	-.0104051	.0109724
educ	.1075228	.0133046	8.08	0.000	.0813715	.133674
exper	.0415623	.0153042	2.72	0.007	.0114805	.0716442
expersq	-.0008152	.0004135	-1.97	0.049	-.001628	-2.33e-06
_cons	-.5333761	.2876625	-1.85	0.064	-1.098802	.0320499

Figure 2: OLS results for a)

Linear regression						
				Number of obs	=	428
				F(5, 422)	=	17.71
				Prob > F	=	0.0000
				R-squared	=	0.1570
				Root MSE	=	.66794

lwage	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
age	-.0006676	.0071394	-0.09	0.926	-.0147009	.0133657
educ	.1095345	.0154197	7.10	0.000	.0792255	.1398434
exper	.0447027	.0183525	2.44	0.015	.0086291	.0807763
expersq	-.0008664	.0004269	-2.03	0.043	-.0017055	-.0000272
imr	.0434592	.2127619	0.20	0.838	-.374746	.4616643
_cons	-.5708801	.3076202	-1.86	0.064	-1.175539	.0337785

Figure 3: Heckman selection correction for c)

Solution.

a) The OLS will be biased due to the self-selection bias.

b) The first step is to run a probit on the probability of working (using the entire sample) with original regressors and the exclusion restrictions. Based on the probit estimates, compute the inverse Mills ratio as

$$IMR = \frac{\phi(x'\hat{\beta})}{\Phi(x'\hat{\beta})}.$$

Then, include this as a regressor in the OLS.

c) Coefficient on the inverse Mills ratio is positive, $\hat{\lambda} = 0.04 > 0$, so

- women with higher unobserved propensity to work also have a higher unobserved productivity at work,
- working women are a positive selection with respect to wages,
- positive correlation of error terms in the selection and outcome equation.

But $\hat{\lambda}$ is not significant, which means that there is no significant evidence for the sample selection bias. In fact, the OLS and Heckman coefficient are very similar.

3 Panel data

Exercise 3.1. Consider the model with individual effects,

$$y_{it} = \alpha_i + x_{it}'\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

Heckman selection model -- two-step estimates				Number of obs	=	753
(regression model with sample selection)				Selected	=	428
				Nonselected	=	325
				Wald chi2(4)	=	51.52
				Prob > chi2	=	0.0000

lwage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
lwage						
age	-.0006676	.0060686	-0.11	0.912	-.0125619	.0112267
educ	.1095345	.0161004	6.80	0.000	.0779782	.1410907
exper	.0447027	.0178719	2.50	0.012	.0096743	.079731
expersq	-.0008664	.0004439	-1.95	0.051	-.0017364	3.70e-06
_cons	-.5708801	.3120663	-1.83	0.067	-1.182519	.0407586
select						
age	-.0528527	.0084772	-6.23	0.000	-.0694678	-.0362376
educ	.1309047	.0252542	5.18	0.000	.0814074	.180402
exper	.1233476	.0187164	6.59	0.000	.0866641	.1600311
expersq	-.0018871	.0006	-3.15	0.002	-.003063	-.0007111
kids05	-.8683285	.1185223	-7.33	0.000	-1.100628	-.636029
kids618	.036005	.0434768	0.83	0.408	-.049208	.1212179
nwifeinc	-.0120237	.0048398	-2.48	0.013	-.0215096	-.0025378
_cons	.2700768	.508593	0.53	0.595	-.7267473	1.266901
/mills						
lambda	.0434592	.1679665	0.26	0.796	-.2857492	.3726675

Figure 4: Heckman selection correction: correct se

with $\beta \in \mathbb{R}^{k\times 1}$.

a) Write down the formula for the within estimator of β.

b) Indicate cases for which not all elements of β could be estimated.

c) Assume that all elements of β could be estimated and specify the formula for the within estimator when $T = 2$. Show that this leads to a regression which could be written as

$$\begin{pmatrix} y_1^\star \\ -y_1^\star \end{pmatrix} = \begin{pmatrix} X_1^\star \\ -X_1^\star \end{pmatrix} \beta + \text{error}.$$

Specify $y_1^\star$ and $X_1^\star$ in terms of y_{it} and x_{it}.

d) Show that the regression in c) yields an estimator that is equivalent to the first-differences estimator.

Exercise 3.2. Modify the random effects model by the addition of a time specific disturbance. So that

$$y_{it} = u_i + v_t + x_{it}'\beta + \varepsilon_{it}, \quad i = 1, \ldots, N, \quad t = 1, \ldots, T$$

where we assume that

- $\mathbb{E}[\varepsilon_{it}] = \mathbb{E}[u_i] = \mathbb{E}[v_t] = 0$,
- $\mathbb{E}[\varepsilon_{it}u_j] = \mathbb{E}[\varepsilon_{it}v_s] = \mathbb{E}[u_i v_t] = 0$, for $\forall\, i, j, t, s$,
- var $[\varepsilon_{it}] = \sigma_\varepsilon^2$, cov $[\varepsilon_{it}, \varepsilon_{js}] = 0$, for $\forall\, i, j, t, s$,
- var $[u_i] = \sigma_u^2$, cov $[u_i, u_j] = 0$, for $\forall\, i, j$,
- var $[v_t] = \sigma_v^2$, cov $[v_t, v_s]$, for $\forall\, t, s$.

Write out the full covariance matrix for a data set with $N = T = 2$.

Exercise 3.3. The two-way fixed effects model is

$$y_{it} = u_i + v_t + x_{it}'\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

where u_i and v_t are the individual-specific and the time-specific intercepts.

a) Show that the within estimator of β, which is the best linear unbiased, could be obtained by applying two within (one-way) transformations of the model.

b) Show that the order of these two withing (one-way) transformations is unimportant. Give an intuitive explanation for this result.

Exercise 3.4. Verify that the infeasible GLS estimator simplifies to the OLS estimator of

$$y_{it}^\star = x_{it}^{\star\prime}\beta + \varepsilon_{it}^\star, \quad i = 1, \dots, N, \quad t = 1, \dots, T$$

where

$$y_{it}^\star := y_{it} - (1-\theta)\bar{y}_i, \quad x_{it}^\star := x_{it} - (1-\theta)\bar{x}_i, \quad \varepsilon_{it}^\star := \theta\alpha_i + \varepsilon_{it} - (1-\theta)\bar{\varepsilon}_i,$$

and

$$\theta^2 := \frac{\sigma_\varepsilon^2}{\sigma_\varepsilon^2 + T\sigma_\alpha^2}.$$

Exercise 3.5. Suppose that the random effects model $y_{it} = x_{it}'\beta + \eta_i + v_{it}$ is to be estimated with a panel in which the groups have different numbers of observations. Let T_i be the number of observations in group i. Show that the pooled least squares estimator is unbiased and consistent despite this complication.

Exercise 3.6. Consider $y_{it} = x_{it}'\beta + \eta_i + v_{it}$, $i = 1, \dots, N$, $t = 1, \dots, T$, where $v_{it} \sim \mathcal{N}(0, \sigma^2)$ and $\beta = 0$. Write out the likelihood for estimating η_i and σ^2, and show that the MLE estimator $\hat{\sigma}^2$ is biased when $T < \infty$.

Exercise 3.7. Consider $y_{it} = \mathbb{1}\{x_{it}\beta + \eta_i + v_{it} \geq 0\}$, where the errors v_{it} have the logisic cdf. Consider $T = 2$, $x_{i1} = 0$ and $x_{i2} = 1$, and show that the sufficient statistic for η_i is $y_{i1} + y_{i2} = 1$, i.e. conditioning on $y_{i1} + y_{i2} = 1$ implies that the MLE does not depend on η_i.

Exercise 3.8. Derive the bias of the OLS estimator for α in a dynamic panel of the form $y_{it} = \alpha y_{it-1} + \eta_i + v_{it}$. Are there any conditions on α that should hold for the estimator to be well-defined?

Exercise 3.9. Consider the panel AR(1) model with individual effects,

$$y_{it} = \alpha y_{it-1} + \eta_i + v_{it}$$

where $\eta_i \sim$ i.i.d.$(0, \sigma_\eta^2)$ and $v_{it} \sim$ i.i.d.$(0, \sigma_v^2)$ are mutually independent, and for all i we have $y_{i0} = 0$. Derive $\text{var}\,[y_{it}]$ for $t = 1, \dots, T$.